

Experiment 7. Case study on simulation of manufacturing system (ARENA)

Aim: Single Machine Production Line

Procedure:

1. Define the System

- Entities: Parts
 - Name: Drill Machine
 - Capacity: 1
- Resource: One drilling machine
- Process: Drilling operation
 - Delay time:
 - Example: Normal(4,1) or Triangular(3,4,6)
- Queue: Before the machine
- Performance measures:
 - WIP (Work-In-Process)
 - Queue length
 - Machine utilization
 - Throughput

2. Build the Model (Basic Process Panel)

Step 1: Create Module (Arrivals)

- Drag Create module
- Name: Parts Arrive
- Define:
 - Arrival type: Random (e.g., Exponential interarrival time)
 - Example: Mean = 5 minutes
- Entity type: Part

Step 2: Process Module (Drilling Operation)

- Drag Process module
- Name: Drilling Process
- Action: Seize–Delay–Release
- Add Resource:

Step 3: Dispose Module (Exit)

- Drag Dispose module
- Name: Parts Exit
- Connect from Process → Dispose

3. Define Resources

- Go to Basic Process → Resource
- Add:
 - Name: Drill Machine
 - Capacity: 1

4. Add Statistics Collection

(a) Queue Statistics

- Automatically tracked in Process module
- Queue name: Drilling Process.Queue

(b) WIP Tracking

Two options:

- Use Number in System (default Arena report)
- Or create a variable:

- Increment at Create
- Decrement at Dispose

(c) Plots (like in image)

- Add Plot module
 - Plot 1: WIP vs Time
 - Plot 2: Queue Length vs Time

5. Run Setup

- Go to Run → Setup
- Replication length:
 - Example: 480 minutes (8 hours)
- Number of replications:
 - Example: 10–30
- Warm-up period (optional):
 - Example: 60 minutes

6. Connect the Flow

Create → Process → Dispose

7. Run the Simulation

- Click Go
- Observe:
 - Animation (parts moving)
 - Queue buildup
 - Machine utilization

8. Analyze Results (Reports)

Check:

- Average WIP
- Average queue length
- Resource utilization

- Throughput (parts/hour)

- Waiting time in queue

Inputs:

Create ? X

Name: Entity Type:

Time Between Arrivals

Type: Value: Units:

Entities per Arrival: Max Arrivals: First Creation:

Action: Priority:

Resources:

Delay Type: Units: Allocation:

Minimum: Value (Most Likely): Maximum:

☒ Report Statistics

Run Setup X

Run Speed | Run Control | Reports

Project Parameters | Replication Parameters | Array Sizes

Number of Replications:

Start Date and Time:

Warm-up Period: Time Units:

Replication Length: Time Units:

Hours Per Day: Base Time Units:

Terminating Condition:

Initialize Between Replications

☒ Statistics ☒ System

Dispose ? X

Name:

State:

Picture ID:

☒ F

Add Copy Delete

Visualization

Size Factor: ☐ Auto Scale ☐ Seize Area

☐ Rotate By Expression

Expression Type:

Basic Process Variables

Entity

Process

Costs

Number In

Number Out

Number In Process

Time

Queue

Resource

Random Distributions

Current Expression:

Plot Expression ? X

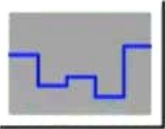
Expression:

Minimum: Maximum:

History Points:

☒ Stepped ☐ Non-Stepped

Color...



Refresh
☐ None
☐ 1/4
☐ 1/2
☐ 3/4
☒ Full

Border
☐ None
☒ Bounding Box
☐ X-Y Axis
☒ Fill Area

Expression Synchronization
☒ Synchronize Min and Max:
 Initial Minimum: Initial Maximum:
☒ Auto Scale

☐ Transparent Background

Title
☒ Use Title
 Percent Height: Vert. Alignment: Horiz. Alignment:
 Title Text: Font...

Axis Labels
☒ X-Labels
☒ Y-Labels
 Font...

Plot ? X

Expressions:
 Add... Edit... Delete

Time Range:

Area... Border... Fill Area...

Refresh
☐ None
☐ 1/4
☐ 1/2
☐ 3/4
☒ Full

Border
☐ None
☒ Bounding Box
☐ X-Y Axis
☒ Fill Area

Expression Synchronization
☒ Synchronize Min and Max:
 Initial Minimum: Initial Maximum:
☒ Auto Scale

☐ Transparent Background

Title
☒ Use Title
 Percent Height: Vert. Alignment: Horiz. Alignment:
 Title Text: Font...

Axis Labels
☒ X-Labels
☒ Y-Labels
 Font...

OK Cancel Help

Clock ? X

Starting Time
 Hour: Min: Sec:

Display
☒ Analog ☐ Digital

Time Format
☒ 12 Hour ☐ 24 Hour

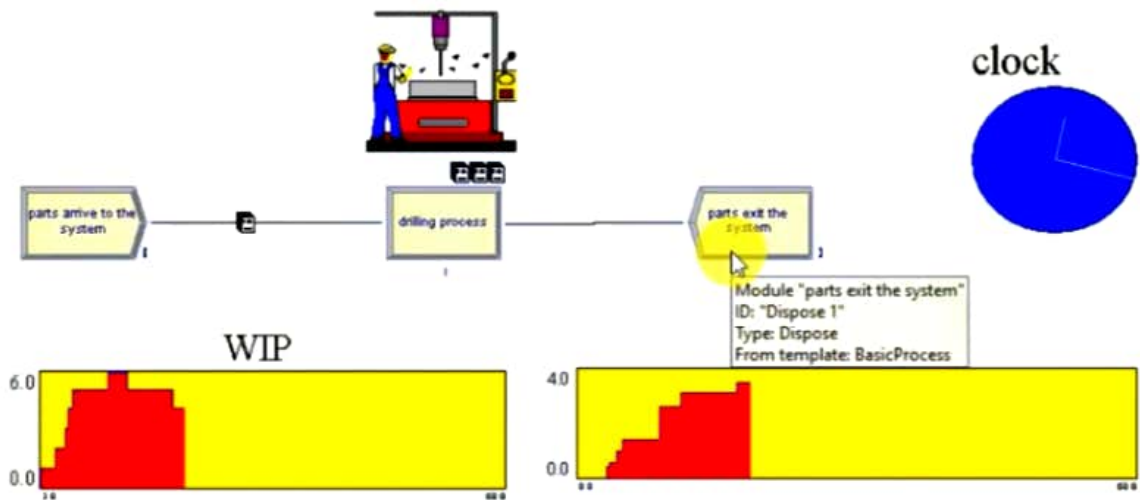
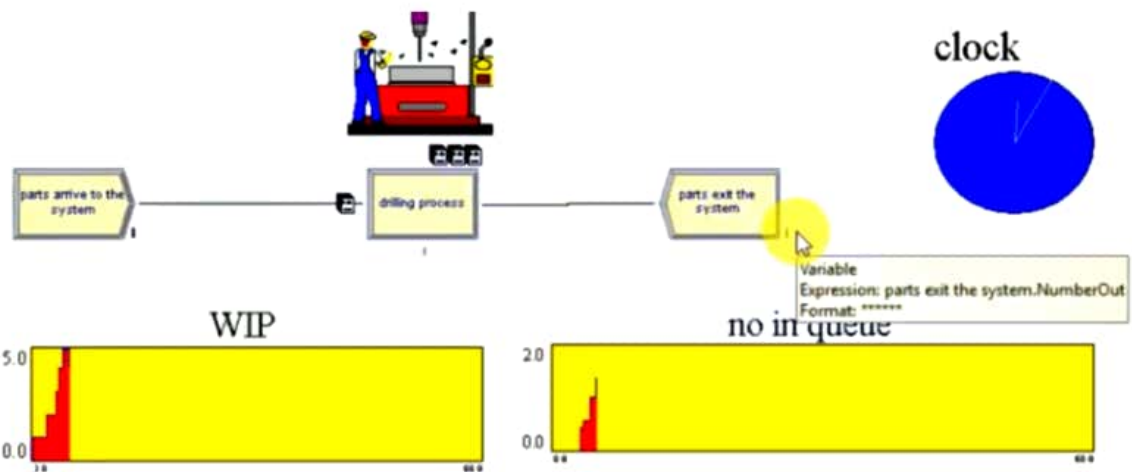
Title
☒ Use Title
 Percent Height: Vert. Alignment: Horiz. Alignment:
 Title Text: Font...

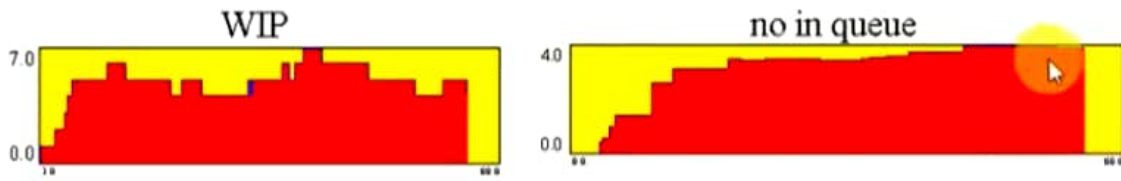
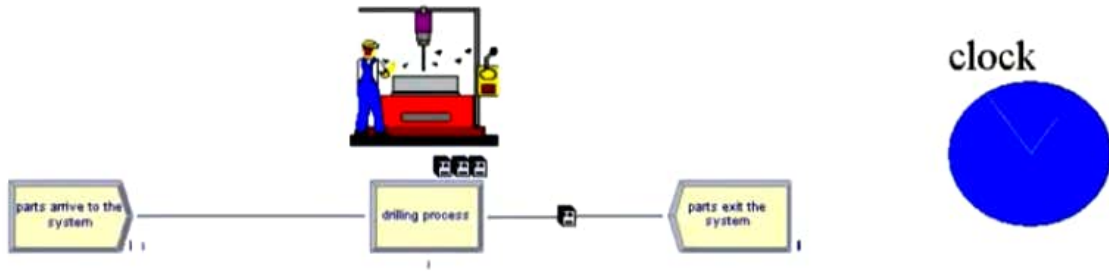
☐ Transparent Background

Area...
Border...
☐ No Border
Hands...

OK Cancel Help

Results:





Unnamed Project

Replications:

Replication 1 Start Time: 0.00 Stop Time: 1,000.00 Time Units: Minute

Queue Detail Summary

Time

drilling process.Queue

Waiting Time
21.24

Other

drilling process.Queue

Number Waiting
4.04

Entity Detail Summary

Time

	NVA Time	Other Time	Total Time	Transfer Time	VA Time
part	0.00	0.00	26.31	0.00	5.07
Total	0.00	0.00	26.31	0.00	5.07

Other

	Number In	Number Out
part	193	188
Total	193	188

Conclusion:

The simulation demonstrates the drilling machine acts as a bottleneck , Queue length and WIP are key indicators of system performance. Small changes in processing time can significantly impact throughput.